Clinical pathway

Upper GI endoscopy → Barrett's esophagus? → Surveillance entry → Carcinoma diagnosed → Histological subtype → Treatment planning

	1 Data audit	2 Representation and model	3 Reliability layer	4 Outcome and evidence
Stage 1 white light endoscopy	cos = 1.00 original / augmented copy **Duplicate audit** every image linked to its augmented copy (1,017 pairs); splits grouped by source image	14 multiscale views (1, 4, 9) → DINOv2 **Frozen foundation features** 14 multiscale views, DINOv2 ViT-B/14; mean pooling, ensemble of 25 members	uncertainty; {BE}; {BE, N}; conformal sets **Calibrated output with deferral** Platt calibration, epistemic deferral, class conditional conformal sets, shift monitor	0.914 internal 0.872 external **Evaluation free of leakage** grouped internal test (204 images); GastroVision, all settings frozen
Stage 2 CT, PET, genomics	G, CT, PET; gate **Missingness audit** does availability alone predict the label? subtype passes; staging excluded	three modalities, any subset CT (34) PET (5) genomics (180) **Fusion with missing modalities** CT, PET and genomic tokens; key padding mask, any subset scored	$p = 0.742$ fusion genomics; modality dropout **Controlled comparison** ablation with genomics only, DeLong test, modality dropout during training	3 × 5 0.870 subtype 199 patients **Nested cross validation** every patient scored by models that never saw that patient

Decision support — **accept** a single confident label · **defer** to the clinician when the conformal set is ambiguous, uncertainty is high or the input lies outside the training scope